

Q. 2

(i) 75° into radians.

$$\frac{75 \times \pi}{180}$$

$$= \frac{5\pi}{12}$$

(ii) $\frac{5\pi}{6}$ into degree

$$\frac{5\pi}{6} \times \frac{180}{\pi}$$

$$150^\circ$$

Q. 3

(i) length of arc = $L = ?$

$$L = r\theta$$

$$L = \frac{6 \times \pi}{3}$$

$$= \frac{6 \times 3.1416}{3}$$

$$= 6.28 \text{ cm}$$

Q2 Area of sector :- $A = ?$

$$A = \frac{1}{2} r^2 \theta$$

$$A = \frac{1}{2} (6)^2 \times \frac{\pi}{3}$$

$$A = \frac{1}{2} \times 18 \times 3.1416$$

$$A = \frac{18 \times 3.1416}{2}$$

$$A = 18.85 \text{ cm}^2$$

Q.4

Classes

C. B

Class-Boundary

Class mark

class interval

Q: 4

(i)

lower class = 15

upper class = 24

(ii) Class boundaries =

$$15 - 0.5 \text{ --- } 24 + 0.5$$

$$14.5 \text{ --- } 24.5$$

(iii) Class mark =

$$x = \frac{15 + 24}{2}$$

$$= \frac{39}{2}$$

$$= 19.5$$

$$= 19.5$$

(iv) Class interval =

$$h = \text{upper limit} - \text{lower limit}$$

$$= 24.5 - 14.5$$

$$= 10$$

Q: 5

(i) mean =

= $\frac{\text{sum of observations}}{\text{N.o of observations}}$

$$= \frac{4 + 6 + 9 + 6 + 10 + 7 + 6 + 12}{8}$$

$$= 7.5$$

Ascending order = 4, 6, 6, 6, 7, 9, 10, 12

Median :-

There are 8 values, so :-

$$\text{Median} = \frac{4^{\text{th}} + 5^{\text{th}} \text{ value}}{2}$$

$$= \frac{6 + 7}{2}$$

$$= \frac{13}{2}$$

$$6.5$$

Mode =

$$\text{mode} = 6$$

Q. 7

(i) Altitudes :- A perpendicular line segment drawn from a vertex of a triangle to the opposite side.

Median :- A line segment joining a vertex of a triangle to the midpoint of the opposite side.

The three altitudes meet at orthocentre

The centroid divides each median in ratio :- 2:1

(1.8)

i)	marks	f	C.B	π	$f\pi$	C.f
	1-10	4	0.5-10.5	$\frac{1+10}{2} = 5.5$	$4 \times 5.5 = 22$	4
	11-20	6	10.5-20.5	$\frac{11+20}{2} = 15.5$	$6 \times 15.5 = 93$	10
	21-30	10	20.5-30.5	$\frac{21+30}{2} = 25.5$	$10 \times 25.5 = 255$	20
	31-40	6	30.5-40.5	$\frac{31+40}{2} = 35.5$	$6 \times 35.5 = 213$	26
	41-50	4	40.5-50.5	$\frac{41+50}{2} = 45.5$	$4 \times 45.5 = 182$	30
		$\Sigma f = 30$			$\Sigma f\pi = 765$	

(i) Arithmetic mean :-

$$\bar{x} = \frac{\Sigma f\pi}{\Sigma f}$$

$$\bar{x} = \frac{765}{30}$$

$$\bar{x} = 25.5$$

(ii) median = $n = 30$, $\frac{n}{2} = 15$

15 observation lies in 21-30 class,
so median group = 21-30

$$L = 20.5$$

$$h = 10$$

$$f = 10$$

$$1 - 10$$

$$\bar{x} = 20.5 + \frac{10}{10} (15 - 10)$$

$$= 20.5 + 5$$

$$\bar{x} = 25.5$$

(iv) mode

modal class = 21-30. } highest f=10

$$L + \left[\frac{f_m - f_1}{2f_m - f_1 - f_2} \right] \times h$$

$$L = 20.5$$

$$f_m = 10$$

$$f_1 = 6$$

$$f_2 = 6$$

$$h = 10$$

$$20.5 + \left(\frac{10 - 6}{2(10) - 6 - 6} \right) \times 10$$

$$20.5 + 5$$

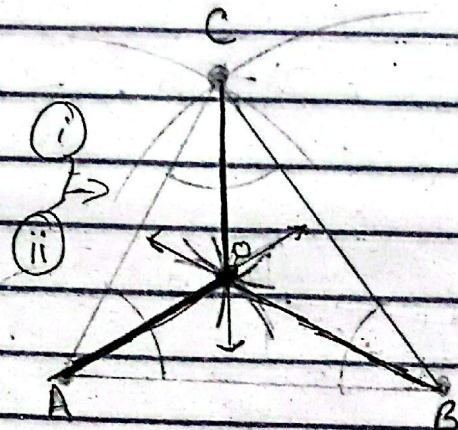
$$= 25.5$$

This means, mean, median and mode are all equal to 25.5, showing that the data is approximately symmetric and centered.

around 25.5 marks.

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(i). $AB = 6\text{cm}$, $BC = 5\text{cm}$ & $AC = 4.5\text{cm}$.



(i) Construction steps:-

① Draw a line segment $AB = 6\text{cm}$.
② With B as a centre and radius 5cm , draw an arc.

③ With A as a centre and radius 4.5cm , draw another arc cutting the first arc at C.

④ Join AC and BC - thus $\triangle ABC$ is constructed.

(ii)

Yes the three angle bisectors are constructed. O is in centre of triangle ABC and lies inside the

MCQs:-

1. C

2. A

3. D

4. B

5. A

6. D

7. B

8. D

9. A

10. B